Exploratory Data Analysis

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1. impute and compute state average. make sure account for survey design by incorporating the weights
2. create a time series database of state policies after June 24, 2022 where the categorical variable has n categories

```
library(haven)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
dat25 <- read_xpt("../data/LLCP2025.XPT")
dat24 <- read_xpt("../data/LLCP2024.XPT")
#dat23 <- read_xpt("../data/LLCP2023.XPT")
dat22 <- read_xpt("../data/LLCP2022.XPT")
#dat21 <- read_xpt("../data/LLCP2021.XPT")
#dat20 <- read_xpt("../data/LLCP2020.XPT")
dat19 <- read_xpt("../data/LLCP2019.XPT")
#dat18 <- read_xpt("../data/LLCP2018.XPT")
dat17 <- read_xpt("../data/LLCP2017.XPT")
```

Exploratory data analysis

variables asked only in 2025, 2024, 2022, 2019, and 2017

NOBCUSE9 = What was main reason for not doing anything to keep you from getting pregnant? PFPPRVN4 = The last time you had sexual intercourse, did you or your partner do anything to keep you from getting pregnant? () FAMILY PLANNING = The last time you had sexual intercourse, what did you or your partner do to keep you from getting pregnant? (lots of response categories!!!)

variables asked in all surveys

SEXVAR = 2 if female

_AGEG5YR = 5-year age groups

EDUCA = What is the highest grade or year of school you completed? (1=no high school, 2=high school, 3=some college,4=college,9=don't know)

MARITAL= marital status (1=married, 2=divorced, 3=widowed, 4=separated, 5=never married,6=A member of an unmarried couple, 9 =refused)

HOME= homeowner status (1=own, 2=rent, 3=other arrangement, 7=don't know, 9=refused)

PREGNANT = current pregnancy status (1=yes, 2=no, 7=don't know, 9=refused)

MENTHLTH = Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? (1-30, 88=none,77=don't know, 99=refused) - consider state-level databases for COVID-19

EMPLOY = employment status (1 = employed for wages, 2 = self-employed, 3 = out of work for 1 year or more, 4 = out of work for less than 1 year, 5 = homemaker, 6 = student, 7 = retired, 8 = unable to work, 9 = refused)

INCOME = income categories (1=less than 15k, 2=15k to 25k, 3= 25k to 35k, 4= 35k to 50k, 5=50k to 100k, 6=100k to 200k, 7=200k or more, 9=don't know)

INSURANCE = Respondents aged 18-64 who have any form of health insurance (1=yes, 2=no, 9=don't know) - consider finding a database that focuses on Medicaid access at the state level and use this as a confounder to address omitted variable bias at the state level

RACE = Race/ethnicity categories (1=nh white, 2=nh black, 3=nh native american, 4=nh asian, 5=nh native hawaiian, 7=nh multiracial,9=don't know)

HLTHCVR1(2017,2019) / PRIMINSR (2022) / PRIMINS2 (2024, 2025) = What is the primary source of your health care coverage? Is it…

subset each survey year to include relevant variables, subsetting the sample to only women

Recode the states in each year since states that didn't collect enough data for a given year weren't included in the dataset

```
sub_dat25 <- dat25 |> select(c("_STATE", "_AGEG5YR","SEXVAR", "PREGNANT",
 ↪  "MENTHLTH", "EMPLOY1","_INCOMG1", "EDUCA", "_HCVU654", "_RACE",
 ↪  "RENTHOM1", "MARITAL", "TYPCNT10", "PRIMINS2")) |> filter(SEXVAR==2) |>
 ↪  select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',
 ↪  EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNT10", "HOME"=RENTHOM1,
 ↪  RACE="_RACE", TYPEINS="PRIMINS2")

state_labels_2025 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
```

```
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `47` = "Tennessee",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico"
)

sub_dat25 <- sub_dat25 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2025[as.character(`_STATE`)]
    )
  )) |>
    mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
sub_dat24 <- dat24 |> select(c("_STATE","_AGEG5YR","SEXVAR","PREGNANT",
 ↪  "MENTHLTH", "EMPLOY1","_INCOMG1", "EDUCA", "_HCVU654","_RACE",
 ↪  "RENTHOM1", "MARITAL", "TYPCNTR9", "PRIMINS2")) |> filter(SEXVAR==2) |>
 ↪  select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU654',
 ↪  EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNTR9", "HOME"=RENTHOM1,
 ↪  RACE="_RACE", TYPEINS="PRIMINS2")
```

```r
state_labels_2024 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `6`  = "California",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
```

```
  `46` = "South Dakota",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico",
  `78` = "Virgin Islands"
)
sub_dat24 <- sub_dat24 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2024[as.character(`_STATE`)]
    )
  )) |> mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
# Roe v. Wade was overturned on June 24, 2022
sub_dat22 <- dat22 |>
 ↪  select(c("_STATE","_AGEG5YR","SEXVAR","IMONTH","IDAY","IYEAR","PREGNANT",
 ↪  "MENTHLTH", "EMPLOY1","_INCOMG1", "MARITAL","EDUCA", "_HCVU652",
 ↪  "_RACE1", "TYPCNTR9", "RENTHOM1", "PRIMINSR")) |> filter(SEXVAR==2) |>
 ↪  select(-SEXVAR) |> rename(INCOME='_INCOMG1', INSURANCE='_HCVU652',
 ↪  EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNTR9", HOME="RENTHOM1",
 ↪  RACE="_RACE1", TYPEINS="PRIMINSR")
state_labels_2022 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `6`  = "California",
  `8`  = "Colorado",
  `9`  = "Connecticut",
```

```
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `47` = "Tennessee",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
```

```
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico",
  `78` = "Virgin Islands"
)
sub_dat22 <- sub_dat22 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
     state_labels_2022[as.character(`_STATE`)]
    )
  ))|> mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
sub_dat19 <- dat19 |> select(c("_STATE","_AGEG5YR","SEXVAR","PREGNANT","MENT
↪  HLTH","EMPLOY1","_INCOMG", "EDUCA", "_HCVU651", "_RACE", "RENTHOM1",
↪  "MARITAL", "TYPCNTR8", "HLTHCVR1")) |> filter(SEXVAR==2) |>
↪  select(-SEXVAR) |> rename(INCOME='_INCOMG', INSURANCE='_HCVU651',
↪  EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNTR8", "HOME"=RENTHOM1,
↪  RACE="_RACE", TYPEINS="HLTHCVR1")
state_labels_2019 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `6`  = "California",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
```

```
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `47` = "Tennessee",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico"
)
sub_dat19 <- sub_dat19 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2019[as.character(`_STATE`)]
    )
  )) |>
    mutate(EDUCA = case_when(
```

```
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
sub_dat17 <- dat17 |> select(c("_STATE", "_AGEG5YR", "SEX","PREGNANT",
 ↪  "MENTHLTH", "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651",  "_RACE",
 ↪  "RENTHOM1", "MARITAL", "TYPCNTR7", "HLTHCVR1")) |> filter(SEX==2) |>
 ↪  select(-SEX) |> rename(INCOME='_INCOMG', INSURANCE='_HCVU651',
 ↪  EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNTR7", "HOME"=RENTHOM1,
 ↪  RACE="_RACE", TYPEINS="HLTHCVR1")
state_labels_2017 <- c(
  `1`  = "Alabama",
  `2`  = "Alaska",
  `4`  = "Arizona",
  `5`  = "Arkansas",
  `6`  = "California",
  `8`  = "Colorado",
  `9`  = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
```

```
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",
  `44` = "Rhode Island",
  `45` = "South Carolina",
  `46` = "South Dakota",
  `47` = "Tennessee",
  `48` = "Texas",
  `49` = "Utah",
  `50` = "Vermont",
  `51` = "Virginia",
  `53` = "Washington",
  `54` = "West Virginia",
  `55` = "Wisconsin",
  `56` = "Wyoming",
  `66` = "Guam",
  `72` = "Puerto Rico"
)
sub_dat17 <- sub_dat17 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2017[as.character(`_STATE`)]
    )
  )) |>
  mutate(EDUCA = case_when(
        EDUCA %in% 1:3 ~ 1, # no high school
        EDUCA == 4 ~ 2, # high school
        EDUCA == 5 ~ 3 ,#Some college
        EDUCA == 6 ~ 4 ,#College graduate
        EDUCA == 9 ~ 9  #Don't know
  ))
```

```
sub_dat23 <- dat23 |> select(c("_STATE","SEXVAR","PREGNANT",
 ↪  "MENTHLTH","EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU653", "_RACE")) |>
 ↪  filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
 ↪  INSURANCE='_HCVU653', EMPLOY='EMPLOY1')
```

```
sub_dat21 <- dat21 |> select(c("_STATE","SEXVAR","PREGNANT",
 ↪  "MENTHLTH","EMPLOY1","_INCOMG1", "EDUCA","_HCVU652", "_RACE")) |>
 ↪  filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
 ↪  INSURANCE='_HCVU652', EMPLOY='EMPLOY1')

sub_dat20 <- dat20 |> select(c("_STATE","SEXVAR","PREGNANT",
 ↪  "MENTHLTH","EMPLOY1", "_INCOMG", "EDUCA",  "_HCVU651", "_RACE")) |>
 ↪  filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG',
 ↪  INSURANCE='_HCVU651', EMPLOY='EMPLOY1')


sub_dat18 <- dat18 |> select(c("_STATE", "SEX1","PREGNANT", "MENTHLTH",
 ↪  "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651",  "_RACE", "RENTHOM1",
 ↪  "MARITAL")) |> filter(SEX1==2) |> select(-SEX1) |>
 ↪  rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1')
```

Variability in insurance type variable over time

```
INSURANCE_TYPE_after_22 = c(
  `1` = "A plan purchased through an employer or union (including plans
   ↪  purchased through another person's employer)",
  `2` = "A private nongovernmental plan that you or another family member
   ↪  buys on your own",
  `3` = "Medicare",
  `4` = "Medigap",
  `5` = "Medicaid",
  `6` = "Children's Health Insurance Program (CHIP)",
  `7` = "Military related health care: TRICARE (CHAMPUS) / VA health care /
   ↪  CHAMP-VA",
  `8` = "Indian Health Service",
  `9` = "State sponsored health plan",
  `10` = "Other government program",
  `88` = "No coverage of any type",
  `77` = "Don't know/Not Sure",
  `99` = "Refused",
  `BLANK` = "Not asked or Missing"
)
```

```
INSURANCE_TYPE_before_22 = c(
  `1` = "A plan purchased through an employer or union; includes plans
  ↪  purchased through another person's employer",
  `2` = "A plan that you or another family member buys on your own",
  `3` = "Medicare",
  `4` = "Medicaid or other state program",
  `5` = "TRICARE (formerly CHAMPUS), VA, or Military",
  `6` = "Alaska Native, Indian Health Service, Tribal Health Services",
  `7` = "Some other source",
  `8` = "None (no coverage)",
  `77` = "Don't know/Not Sure",
  `99` = "Refused",
  `BLANK` = "Not asked or Missing"
)

library(dplyr)
library(ggplot2)
library(viridis)
```

```
Loading required package: viridisLite
```

```
insurance_type_plot <- function(var, data, label_map, plot_title) {

  factor_var <- paste0(var, "_factor")

  data[[factor_var]] <- factor(
    as.character(data[[var]]),
    levels = names(label_map)
  )

  legend_labels <- function(x) {
    x <- as.character(x)
    value_labels <- unname(label_map[x])

    # Use the numeric code if a label is unavailable
    value_labels[is.na(value_labels)] <- x[is.na(value_labels)]

    paste0(x, " = ", value_labels)
  }

  ggplot(
```

```
    data,
    aes(
      x = .data[[factor_var]],
      fill = .data[[factor_var]]
    )
  ) +
    geom_bar(color = "white") +
    scale_fill_viridis_d(
  name = var,
  labels = legend_labels,
  na.value = "grey70",
  na.translate = TRUE
) +
    theme_minimal() +
    theme(
      axis.text.x = element_text(
        angle = 45,
        hjust = 1
      ),
      legend.position = "right",
      legend.text = element_text(size = 9)
    ) +
    labs(
      title = plot_title,
      x = "Numeric response code",
      y = "Count"
    )
}


# Create output folder
output_dir <- "plots"
dir.create(output_dir, showWarnings = FALSE, recursive = TRUE)

# Create plots
p_2017 <- insurance_type_plot(
  var = "TYPEINS",
  data = sub_dat17 |>
    rename(age = "_AGEG5YR") |>
    filter(age < 7),
  label_map = INSURANCE_TYPE_before_22,
  plot_title = "Health Insurance Type, BRFSS 2017"
)
```

```r
p_2019 <- insurance_type_plot(
  var = "TYPEINS",
  data = sub_dat19 |>
    rename(age = "_AGEG5YR") |>
    filter(age < 7),
  label_map = INSURANCE_TYPE_before_22,
  plot_title = "Health Insurance Type, BRFSS 2019"
)

p_2022 <- insurance_type_plot(
  var = "TYPEINS",
  data = sub_dat22 |>
    rename(age = "_AGEG5YR") |>
    filter(age < 7),
  label_map = INSURANCE_TYPE_after_22,
  plot_title = "Health Insurance Type, BRFSS 2022"
)

p_2024 <- insurance_type_plot(
  var = "TYPEINS",
  data = sub_dat24 |>
    rename(age = "_AGEG5YR") |>
    filter(age < 7),
  label_map = INSURANCE_TYPE_after_22,
  plot_title = "Health Insurance Type, BRFSS 2024"
)

p_2025 <- insurance_type_plot(
  var = "TYPEINS",
  data = sub_dat25 |>
    rename(age = "_AGEG5YR") |>
    filter(age < 7),
  label_map = INSURANCE_TYPE_after_22,
  plot_title = "Health Insurance Type, BRFSS 2025"
)
ggsave(
  filename = file.path(output_dir, "health_insurance_type_2017.png"),
  plot = p_2017,
  width = 14,
  height = 9,
  units = "in",
  dpi = 300
```

```
)

ggsave(
  filename = file.path(output_dir, "health_insurance_type_2019.png"),
  plot = p_2019,
  width = 14,
  height = 9,
  units = "in",
  dpi = 300
)

ggsave(
  filename = file.path(output_dir, "health_insurance_type_2022.png"),
  plot = p_2022,
  width = 14,
  height = 9,
  units = "in",
  dpi = 300
)

ggsave(
  filename = file.path(output_dir, "health_insurance_type_2024.png"),
  plot = p_2024,
  width = 14,
  height = 9,
  units = "in",
  dpi = 300
)

ggsave(
  filename = file.path(output_dir, "health_insurance_type_2025.png"),
  plot = p_2025,
  width = 14,
  height = 9,
  units = "in",
  dpi = 300
)
```

```
library(janitor)
```

```
Attaching package: 'janitor'
```

```
The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
# level of missingness in insurance type by year
cat("Health Insurance Type, BRFSS 2017")
```

```
Health Insurance Type, BRFSS 2017
```

```
sub_dat17 |> rename(age = "_AGEG5YR") |>
    filter(age < 7) |> tabyl(TYPEINS)
```

```
 TYPEINS     n      percent valid_percent
       1  3831 0.0476931504   0.566129747
       2   757 0.0094240968   0.111866411
       3   490 0.0061001419   0.072410226
       4  1237 0.0153997460   0.182798877
       5   121 0.0015063616   0.017880893
       6    10 0.0001244927   0.001477760
       7   219 0.0027263900   0.032362938
       8    13 0.0001618405   0.001921088
      77    69 0.0008589996   0.010196542
      99    20 0.0002489854   0.002955519
      NA 73559 0.9157557951            NA
```

```
cat("Health Insurance Type, BRFSS 2019")
```

```
Health Insurance Type, BRFSS 2019
```

```
sub_dat19 |> rename(age = "_AGEG5YR") |>
    filter(age < 7) |> tabyl(TYPEINS)
```

```
 TYPEINS     n     percent valid_percent
       1  8500 0.116007697   0.572043879
       2  1415 0.019311870   0.095228481
       3   895 0.012214928   0.060232856
       4  2628 0.035866850   0.176862508
       5   290 0.003957910   0.019516791
```

```
  6    29 0.000395791   0.001951679
  7   510 0.006960462   0.034322633
  8   313 0.004271813   0.021064675
 77   191 0.002606761   0.012854162
 99    88 0.001201021   0.005922337
 NA 58412 0.797204897            NA
```

```
cat("Health Insurance Type, BRFSS 2022")
```

```
Health Insurance Type, BRFSS 2022
```

```
sub_dat22 |> rename(age = "_AGEG5YR") |>
    filter(age < 7) |> tabyl(TYPEINS)
```

```
 TYPEINS     n      percent valid_percent
       1 41452 5.164329e-01   0.516439295
       2  7360 9.169511e-02   0.091696256
       3  3138 3.909501e-02   0.039095496
       4    93 1.158647e-03   0.001158662
       5 11009 1.371565e-01   0.137158164
       6    87 1.083896e-03   0.001083910
       7  1856 2.312312e-02   0.023123404
       8   346 4.310667e-03   0.004310721
       9  3452 4.300700e-02   0.043007538
      10  2143 2.669873e-02   0.026699059
      77  2116 2.636235e-02   0.026362674
      88  6406 7.980963e-02   0.079810627
      99   807 1.005407e-02   0.010054195
      NA     1 1.245858e-05            NA
```

```
cat("Health Insurance Type, BRFSS 2024")
```

```
Health Insurance Type, BRFSS 2024
```

```
sub_dat24 |> rename(age = "_AGEG5YR") |>
    filter(age < 7) |> tabyl(TYPEINS)
```

TYPEINS	n	percent
1	39914	0.4836653580
2	8076	0.0978624400
3	3032	0.0367408269
4	56	0.0006785905
5	12540	0.1519557947
6	60	0.0007270612
7	1807	0.0218966604
8	350	0.0042411904
9	3482	0.0421937861
10	2060	0.0249624352
77	2765	0.0335054045
88	7687	0.0931486598
99	695	0.0084217924

```
cat("Health Insurance Type, BRFSS 2025")
```

```
Health Insurance Type, BRFSS 2025
```

```
sub_dat25 |> rename(age = "_AGEG5YR") |>
    filter(age < 7) |> tabyl(TYPEINS)
```

TYPEINS	n	percent
1	29744	0.4794017149
2	6175	0.0995261427
3	2315	0.0373122300
4	40	0.0006447038
5	9808	0.1580813616
6	54	0.0008703501
7	1422	0.0229192186
8	244	0.0039326929
9	2370	0.0381986977
10	1363	0.0219682806
77	2212	0.0356521179
88	5759	0.0928212236
99	538	0.0086712656

merge relevant questionnaire years

```r
data_list <- list(
  `2017` = sub_dat17,
  `2019` = sub_dat19,
  `2022` = sub_dat22,
  `2024` = sub_dat24,
  `2025` = sub_dat25
)

vars_to_keep <- c(
  "_STATE",
  "PREGNANT",
  "_AGEG5YR",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE",
  "MARITAL",
  "HOME",
  "TYPEINS",
  "FAMILY_PLANNING",
  "IYEAR",
  "IMONTH",
  "IDAY"
)

analysis_data <- purrr::imap_dfr(
  data_list,
  function(dataset, year) {

    year <- as.integer(year)

    dataset <- dataset |>
      dplyr::select(dplyr::any_of(vars_to_keep)) |>
      dplyr::mutate(
        YEAR = year
      )

    if (year == 2022) {
      dataset <- dataset |>
        dplyr::mutate(
```

```
          IYEAR = as.integer(as.character(IYEAR)),
          IMONTH = as.integer(as.character(IMONTH)),
          IDAY = as.integer(as.character(IDAY)),

          date = as.Date(
            sprintf("%04d-%02d-%02d", IYEAR, IMONTH, IDAY)
          ),

          post_ban = dplyr::case_when(
            date < as.Date("2022-06-24") ~ 0L,
            date >= as.Date("2022-06-24") ~ 1L,
            TRUE ~ NA_integer_
          )
        )
    } else {
      dataset <- dataset |>
        dplyr::mutate(
          date = as.Date(NA),

          post_ban = dplyr::case_when(
            YEAR <= 2021 ~ 0L,
            YEAR >= 2023 ~ 1L,
            TRUE ~ NA_integer_
          )
        )
    }

    dataset
  }
) |>
  dplyr::rename(STATE = `_STATE`) |> select(-c(IYEAR, IMONTH, IDAY, date))


# check that post ban was correctly created
analysis_data |> filter(YEAR == 2025) |> janitor::tabyl(post_ban)
```

```
 post_ban      n percent
        1 187696       1
```

```
analysis_data |> filter(YEAR == 2024) |> janitor::tabyl(post_ban)
```

```
 post_ban      n percent
```

```
        1 240183       1
```

```
analysis_data |> filter(YEAR == 2022) |> janitor::tabyl(post_ban)
```

```
 post_ban      n      percent valid_percent
        0  96964 4.110508e-01      0.411056
        1 138926 5.889365e-01      0.588944
       NA      3 1.271763e-05            NA
```

```
analysis_data |> filter(YEAR == 2019) |> janitor::tabyl(post_ban)
```

```
 post_ban      n percent
        0 228419       1
```

```
analysis_data |> filter(YEAR == 2017) |> janitor::tabyl(post_ban)
```

```
 post_ban      n percent
        0 251007       1
```

create U.S. census regions

```
library(tibble)

census_region_lookup <- tribble(
  ~STATE, ~REGION, ~DIVISION,

  # Northeast
  "Connecticut",          "Northeast", "New England",
  "Maine",                "Northeast", "New England",
  "Massachusetts",        "Northeast", "New England",
  "New Hampshire",        "Northeast", "New England",
  "Rhode Island",         "Northeast", "New England",
  "Vermont",              "Northeast", "New England",
  "New Jersey",           "Northeast", "Middle Atlantic",
  "New York",             "Northeast", "Middle Atlantic",
  "Pennsylvania",         "Northeast", "Middle Atlantic",
```

```
# Midwest
"Illinois",             "Midwest", "East North Central",
"Indiana",              "Midwest", "East North Central",
"Michigan",             "Midwest", "East North Central",
"Ohio",                 "Midwest", "East North Central",
"Wisconsin",            "Midwest", "East North Central",
"Iowa",                 "Midwest", "West North Central",
"Kansas",               "Midwest", "West North Central",
"Minnesota",            "Midwest", "West North Central",
"Missouri",             "Midwest", "West North Central",
"Nebraska",             "Midwest", "West North Central",
"North Dakota",         "Midwest", "West North Central",
"South Dakota",         "Midwest", "West North Central",

# South
"Delaware",             "South", "South Atlantic",
"District of Columbia", "South", "South Atlantic",
"Florida",              "South", "South Atlantic",
"Georgia",              "South", "South Atlantic",
"Maryland",             "South", "South Atlantic",
"North Carolina",       "South", "South Atlantic",
"South Carolina",       "South", "South Atlantic",
"Virginia",             "South", "South Atlantic",
"West Virginia",        "South", "South Atlantic",
"Alabama",              "South", "East South Central",
"Kentucky",             "South", "East South Central",
"Mississippi",          "South", "East South Central",
"Tennessee",            "South", "East South Central",
"Arkansas",             "South", "West South Central",
"Louisiana",            "South", "West South Central",
"Oklahoma",             "South", "West South Central",
"Texas",                "South", "West South Central",

# West
"Arizona",              "West", "Mountain",
"Colorado",             "West", "Mountain",
"Idaho",                "West", "Mountain",
"Montana",              "West", "Mountain",
"Nevada",               "West", "Mountain",
"New Mexico",           "West", "Mountain",
"Utah",                 "West", "Mountain",
"Wyoming",              "West", "Mountain",
"Alaska",               "West", "Pacific",
```

```
  "California",           "West", "Pacific",
  "Hawaii",               "West", "Pacific",
  "Oregon",               "West", "Pacific",
  "Washington",           "West", "Pacific",

  # Territories: not assigned to a Census region/division
  "Guam",                 NA_character_, NA_character_,
  "Puerto Rico",          NA_character_, NA_character_,
  "Virgin Islands",       NA_character_, NA_character_
)

# remove Guam, Puerto Rico, and Virgin Islands from analytic sample
analysis_data_v2 <- analysis_data |>
  dplyr::left_join(
    census_region_lookup,
    by = "STATE"
  ) |> filter(!is.na(REGION)) |> rename(AGEG5YR = "_AGEG5YR")
```

```
variable_labels <- list(

  AGEG5YR = c(
  `1`  = "18 to 24 years",
  `2`  = "25 to 29 years",
  `3`  = "30 to 34 years",
  `4`  = "35 to 39 years",
  `5`  = "40 to 44 years",
  `6`  = "45 to 49 years",
  `7`  = "50 to 54 years",
  `8`  = "55 to 59 years",
  `9`  = "60 to 64 years",
  `10` = "65 to 69 years",
  `11` = "70 to 74 years",
  `12` = "75 to 79 years",
  `13` = "80 years or older",
  `14` = "Don't know/Refused/Missing"
),

  MARITAL = c(
    `1` = "Married",
    `2` = "Divorced",
    `3` = "Widowed",
    `4` = "Separated",
```

```
  `5` = "Never married",
  `6` = "Member of an unmarried couple",
  `9` = "Refused"
),

HOME = c(
  `1` = "Own",
  `2` = "Rent",
  `3` = "Other arrangement",
  `7` = "Don't know",
  `9` = "Refused"
),

PREGNANT = c(
  `1` = "Yes",
  `2` = "No",
  `7` = "Don't know",
  `9` = "Refused"
),

MENTHLTH = c(
  setNames(
    paste(1:30, ifelse(1:30 == 1, "day", "days")),
    as.character(1:30)
  ),
  `77` = "Don't know",
  `88` = "None",
  `99` = "Refused"
),

EMPLOY = c(
  `1` = "Employed for wages",
  `2` = "Self-employed",
  `3` = "Out of work for 1 year or more",
  `4` = "Out of work for less than 1 year",
  `5` = "Homemaker",
  `6` = "Student",
  `7` = "Retired",
  `8` = "Unable to work",
  `9` = "Refused"
),

INCOME = c(
```

```
    `1` = "Less than $15,000",
    `2` = "$15,000 to $24,999",
    `3` = "$25,000 to $34,999",
    `4` = "$35,000 to $49,999",
    `5` = "$50,000 or more", # 2017, 2019 data only has $50k or more so align
    ↪  other years to this format
    `6` = "$50,000 or more",
    `7` = "$50,000 or more",
    `9` = "Don't know"
  ),

  EDUCA = c(
    `1` = "No high school",
    `2` = "High school",
    `3` = "Some college",
    `4` = "College graduate",
    `9` = "Don't know"
  ),

  INSURANCE = c(
    `1` = "Yes",
    `2` = "No",
    `9` = "Don't know"
  ),

  RACE = c(
    `1` = "Non-Hispanic White",
    `2` = "Non-Hispanic Black",
    `3` = "Non-Hispanic Native American",
    `4` = "Non-Hispanic Asian",
    `5` = "Non-Hispanic Native Hawaiian",
    `6` = "Other race, non-Hispanic",
    `7` = "Non-Hispanic Multiracial",
    `8` = "Hispanic",
    `9` = "Don't know"
  )
)

variables_to_summarize <- c(
  "MARITAL",
  "HOME",
  "PREGNANT",
  "MENTHLTH",
```

```
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE"
)
```

Assess Missingness by Age Group

6 = Age 45 to 49 (menopause starts around this age range) Pregnancy Status `1` = "Yes", `2` = "No", `7` = "Don't know", `9` = "Refused"

`INSURANCE` = Respondents aged 18-64 who have any form of health insurance (1=yes, 2=no, 9=don't know)

Women who are 50 years old or older (code 7 and above) did not answer the pregnancy question.

```
analysis_data_v2 |>
  count(AGEG5YR, PREGNANT) |>
  group_by(AGEG5YR) |>
  mutate(percent = round(100 *n /sum(n),3)) |>
  ungroup()
```

```
# A tibble: 45 x 4
   AGEG5YR PREGNANT     n percent
     <dbl>    <dbl> <int>   <dbl>
 1       1        1  2254   4.01
 2       1        2 51880  92.3
 3       1        7   162   0.288
 4       1        9    81   0.144
 5       1       NA  1813   3.23
 6       2        1  2872   5.77
 7       2        2 45056  90.5
 8       2        7   184   0.37
 9       2        9   115   0.231
10       2       NA  1558   3.13
# i 35 more rows
```

```r
analysis_data_v2 |>
  count(AGEG5YR, INSURANCE) |>
  group_by(AGEG5YR) |>
  mutate(percent = round(100 *n /sum(n),3)) |>
  ungroup()
```

```
# A tibble: 32 x 4
   AGEG5YR INSURANCE     n percent
     <dbl>     <dbl> <int>   <dbl>
 1       1         1 45387   80.8
 2       1         2  6234   11.1
 3       1         9  4569    8.13
 4       2         1 42106   84.6
 5       2         2  6504   13.1
 6       2         9  1175    2.36
 7       3         1 50889   86.5
 8       3         2  6917   11.8
 9       3         9  1035    1.76
10       4         1 57814   88.1
# i 22 more rows
```

U.S. Population Characteristics Over Time

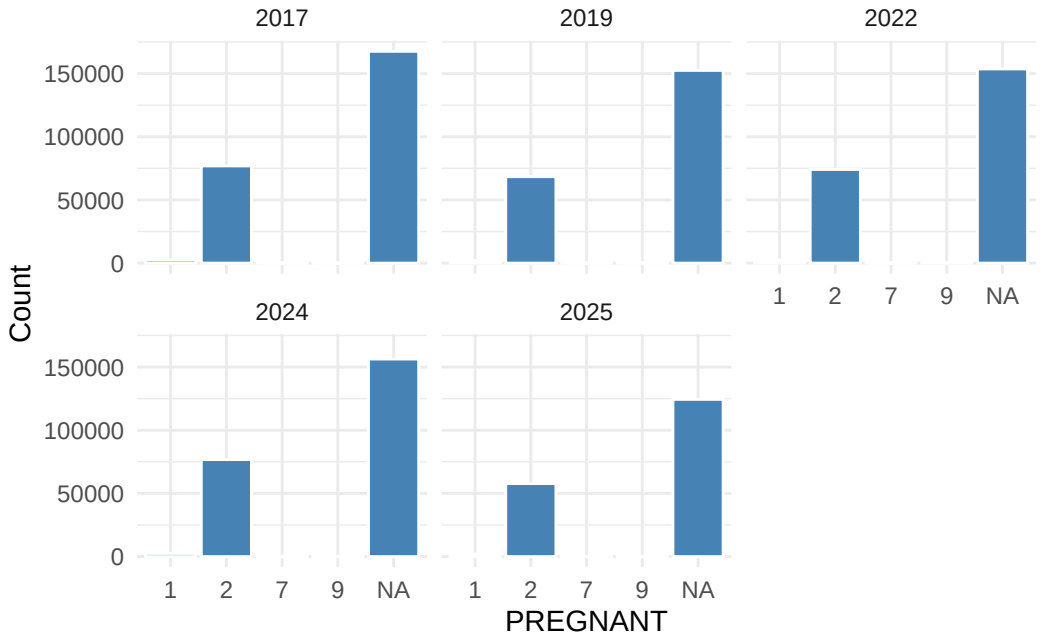
This sample consist of women of all ages.

```r
# categorical variables
vars_to_plot <- setdiff(
  names(analysis_data_v2),
  c("STATE", "YEAR", "post_ban", "DIVISION", "MENTHLTH", "REGION")
)
lapply(vars_to_plot, function(var){
  label_map <- variable_labels[[var]]

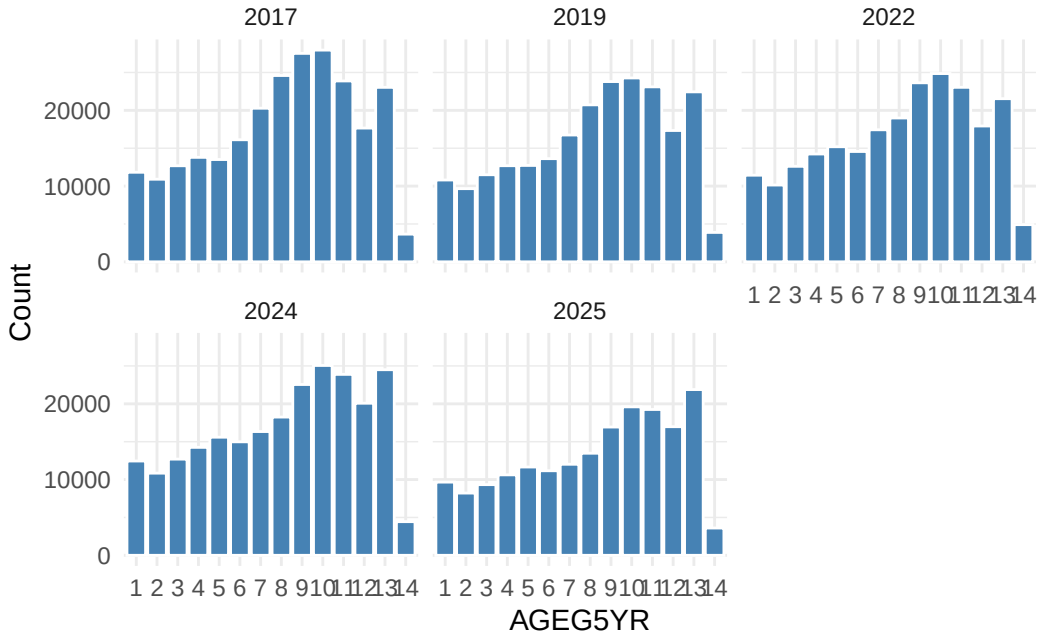
  ggplot(
  analysis_data_v2,
  aes(x = factor(.data[[var]]))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
```

```
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    x = var,
    y = "Count"
  )
})
```

```
[[1]]
```



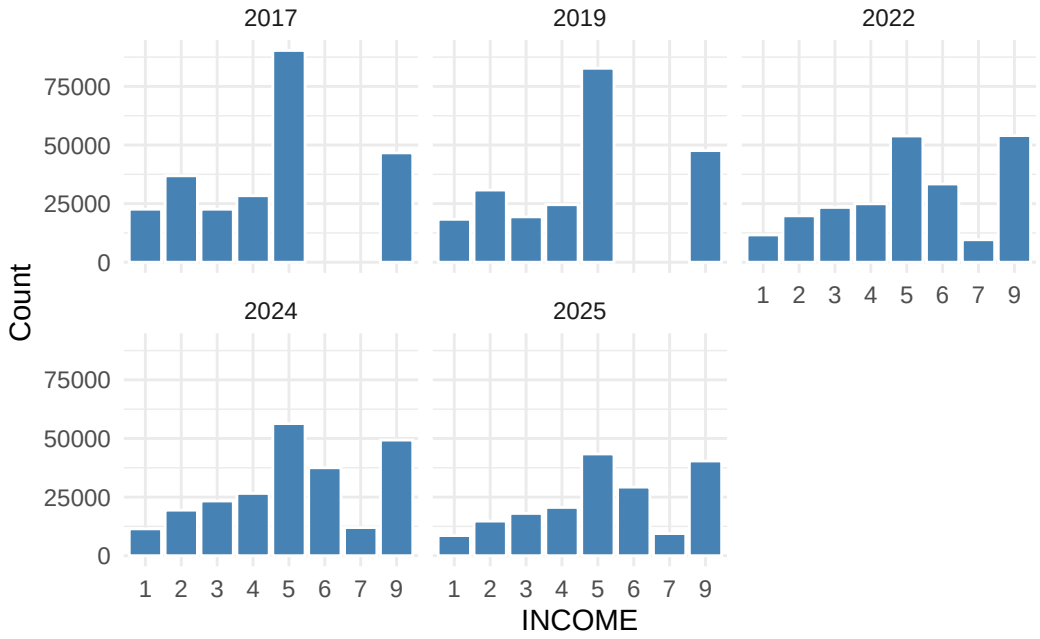

```
[[2]]
```


2017
2019
2022
2024
2025
Count
20000
10000
0
1 2 3 4 5 6 7 8 9 10 11 12 13 14
AGEG5YR

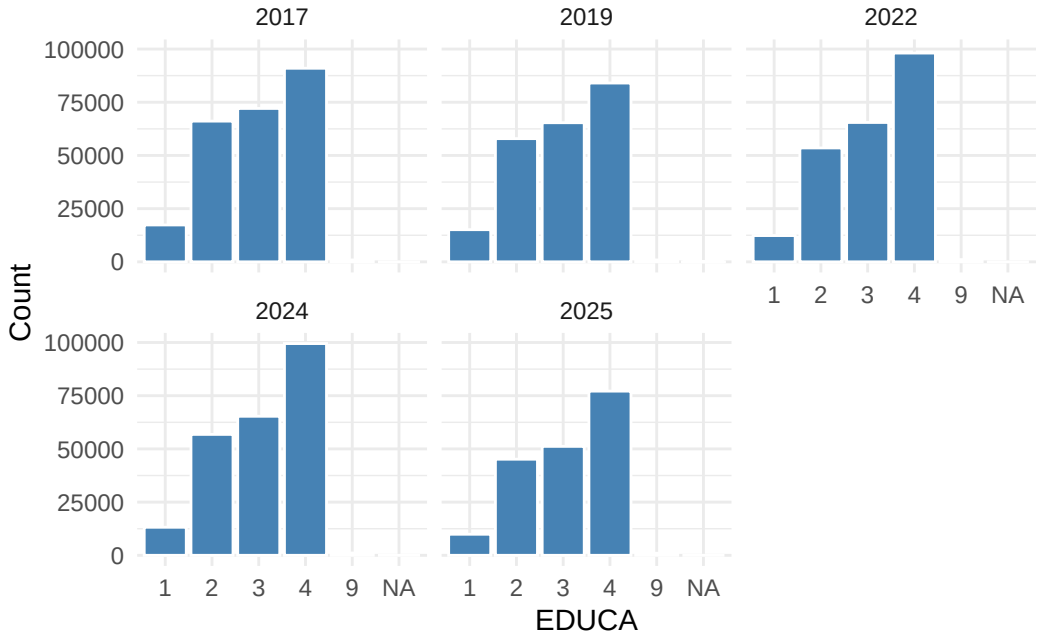
[[3]]


2017
2019
2022
2024
2025
Count
75000
50000
25000
0
1 2 3 4 5 6 7 8 9 NA
EMPLOY

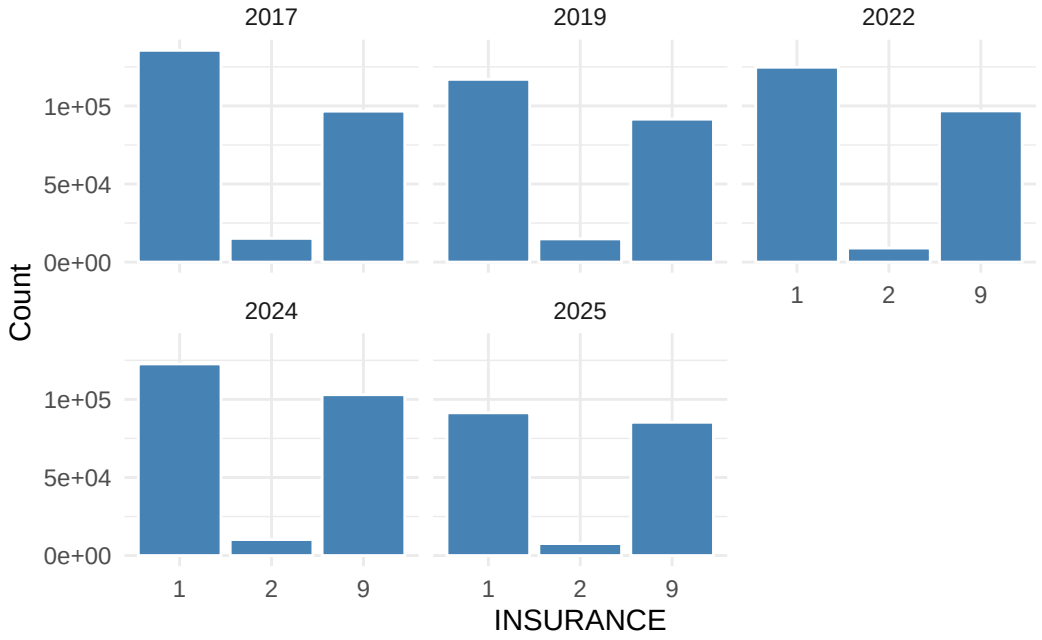
[[4]]


2017
2019
2022
2024
2025
Count
75000
50000
25000
0
1 2 3 4 5 6 7 9
INCOME

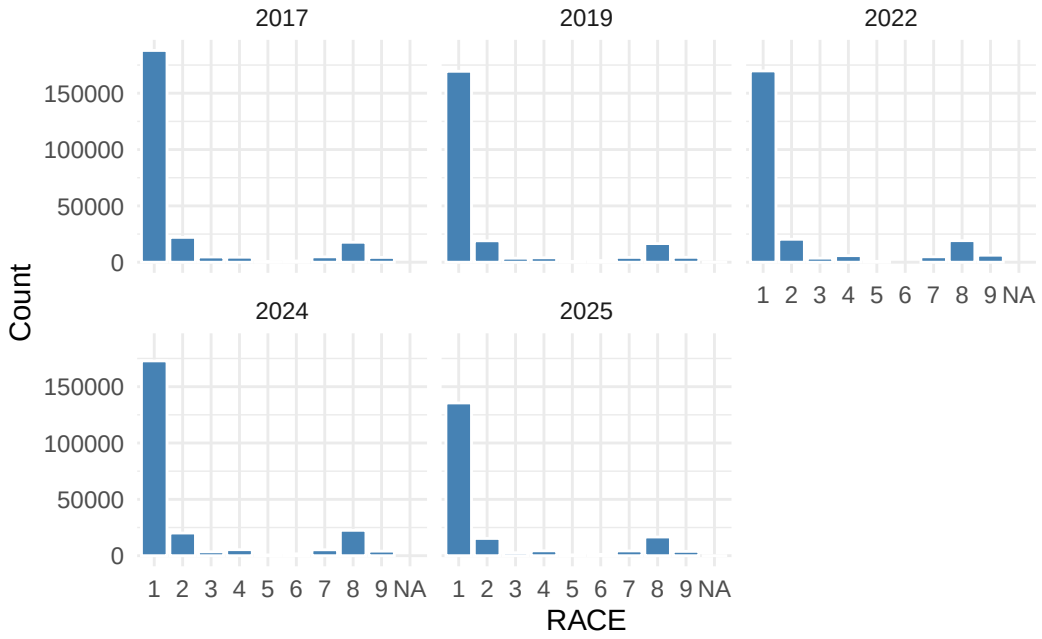
[[5]]


2017
2019
2022
2024
2025
Count
100000
75000
50000
25000
0
1
2
3
4
9
NA
EDUCA

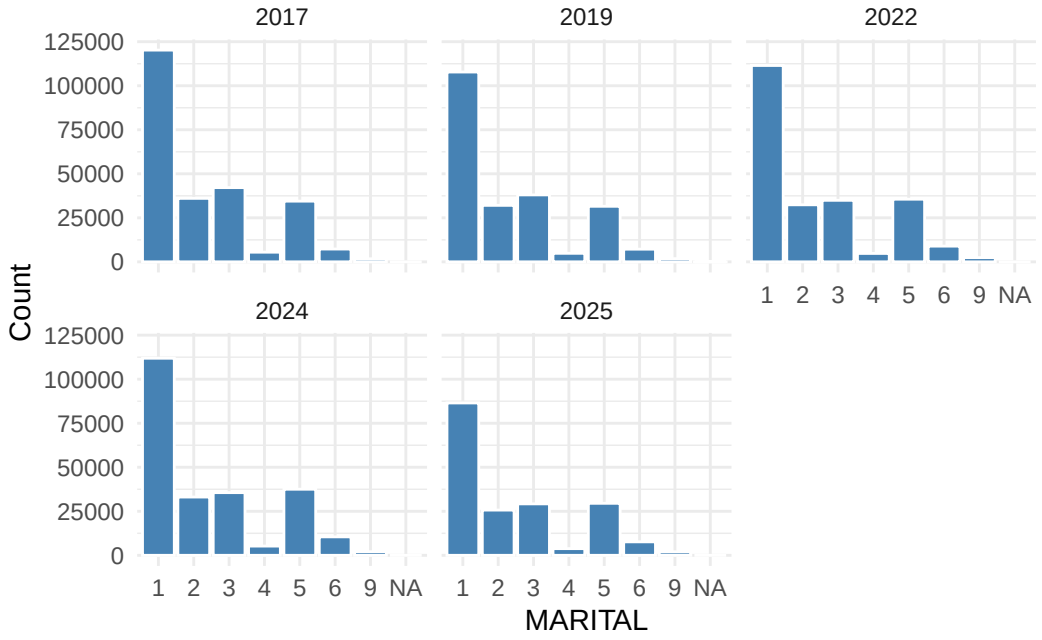
[[6]]


2017
2019
2022
2024
2025
Count
1e+05
5e+04
0e+00
1
2
9
INSURANCE

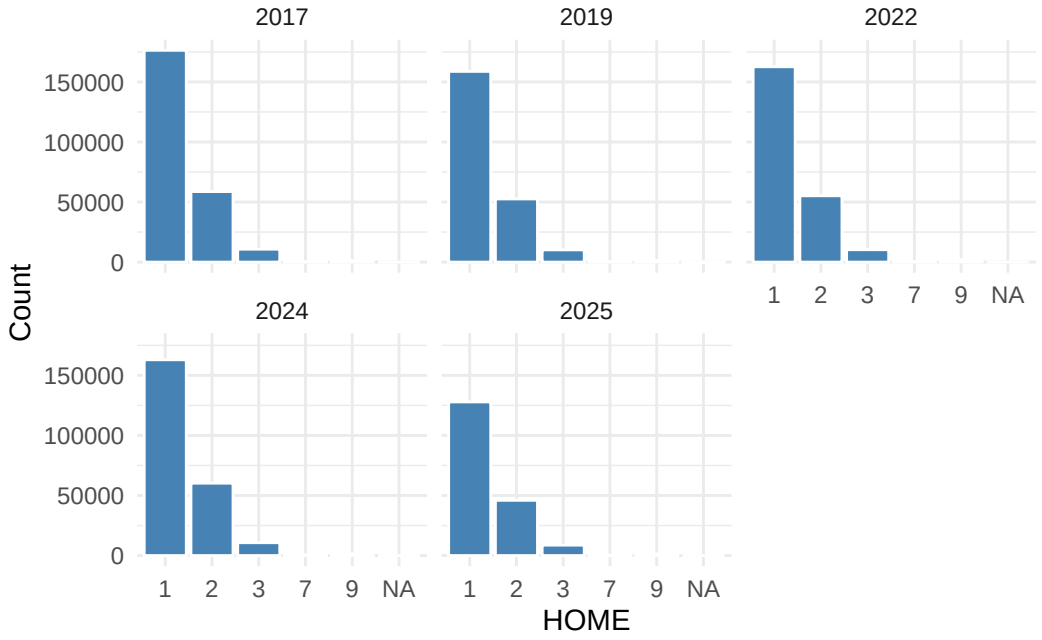
[[7]]


2017
2019
2022
2024
2025
Count
150000
100000
50000
0
1 2 3 4 5 6 7 8 9 NA
RACE

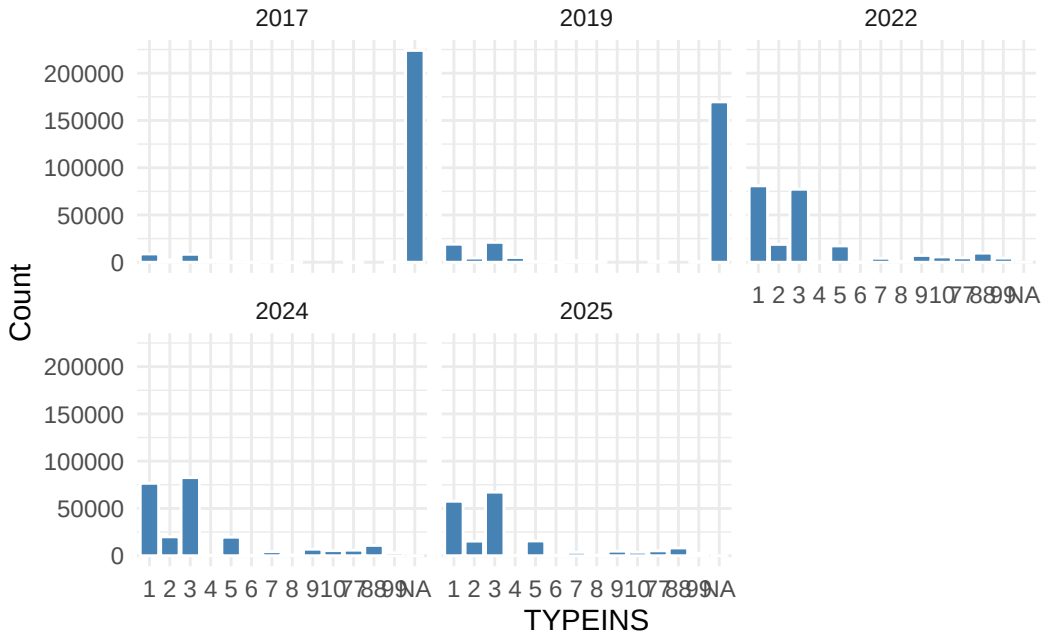
[[8]]


2017
2019
2022
2024
2025
Count
125000
100000
75000
50000
25000
0
1 2 3 4 5 6 9 NA
MARITAL

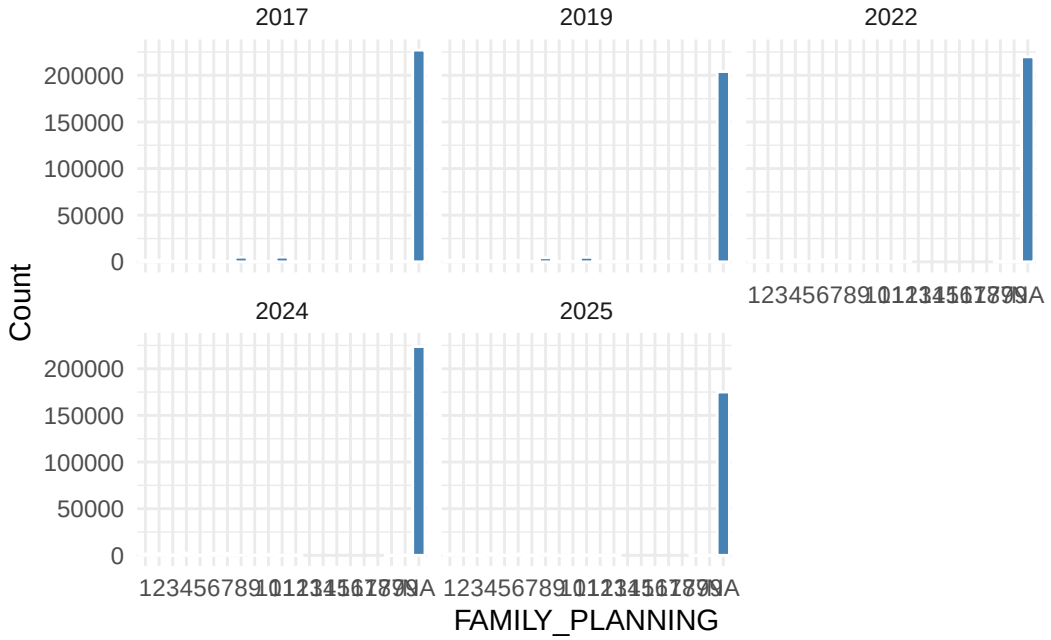
[[9]]


2017
2019
2022
2024
2025
Count
150000
100000
50000
0
1 2 3 7 9 NA
HOME

```
[[10]]
```

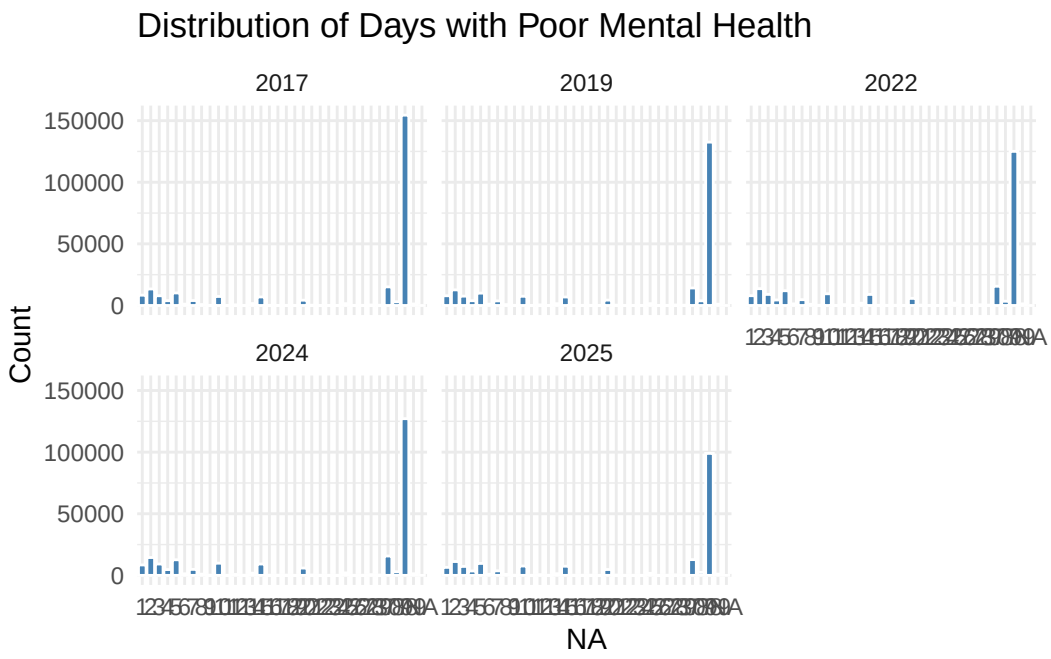

2017
2019
2022
2024
2025
Count
200000
150000
100000
50000
0
1 2 3 4 5 6 7 8 9 10 77 88 99 NA
TYPEINS

```
[[11]]
```




```
# numerical variable
  ggplot(
  analysis_data_v2 ,
  aes(x = factor(MENTHLTH))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    title = "Distribution of Days with Poor Mental Health",
    x = var,
    y = "Count"
  )
```

```
Warning in label(labels[[nm]] %||% ""): NAs introduced by coercion
```




```
get_descriptive_stats_over_entire_period <- function(data, var) {

  tab <- data |>
    janitor::tabyl(
      {{ var }},
      show_na = TRUE
    )

  # The first column is the categorical variable
  category_col <- names(tab)[1]

  total_n <- sum(tab$n)

  valid_n <- sum(
    tab$n[!is.na(tab[[category_col]])]
  )

  tab |>
    dplyr::mutate(
      percent = 100 * n / total_n,
      valid_percent = dplyr::if_else(
        valid_n > 0 & !is.na(.data[[category_col]]),
        100 * n / valid_n,
```

```
        NA_real_
      )
    )
}
get_descriptive_stats_over_entire_period(analysis_data,PREGNANT)

get_descriptive_stats_over_entire_period(analysis_data,MENTHLTH)
get_descriptive_stats_over_entire_period(analysis_data,EMPLOY)
get_descriptive_stats_over_entire_period(analysis_data,INCOME)
get_descriptive_stats_over_entire_period(analysis_data,EDUCA)
get_descriptive_stats_over_entire_period(analysis_data,INSURANCE)
```

Summary Table of Count and Percentages

```
library(dplyr)
library(tidyr)
library(purrr)

summarize_categorical <- function(
    data,
    group_vars,
    response_vars,
    variable_labels
) {

  group_vars <- as.character(group_vars)
  response_vars <- as.character(response_vars)

  required_vars <- c(group_vars, response_vars)
  missing_vars <- setdiff(required_vars, names(data))

  if (length(missing_vars) > 0) {
    stop(
      "These variables are not in the dataset: ",
      paste(missing_vars, collapse = ", ")
    )
  }

  # Function to retrieve labels
  lookup_label <- function(variable, code) {
```

```
  # Actual NA values
  if (is.na(code) || code == "") {
    return("Missing")
  }

  labels <- variable_labels[[variable]]

  # No label vector exists for the variable
  if (is.null(labels)) {
    return(paste("Unlabeled variable:", variable))
  }

  label <- unname(labels[as.character(code)])

  # Code does not exist in the label vector
  if (length(label) == 0 || is.na(label)) {
    return(paste("Unlabeled code:", code))
  }

  label
}

output_data <- data |>
  select(all_of(required_vars))

# Create labeled versions of subgroup variables
subgroup_label_vars <- character(0)

for (group_var in group_vars) {

  if (group_var %in% names(variable_labels)) {

    label_var <- paste0(group_var, "_LABEL")

    output_data[[label_var]] <- purrr::map_chr(
      as.character(output_data[[group_var]]),
      ~ lookup_label(group_var, .x)
    )

    subgroup_label_vars <- c(subgroup_label_vars, label_var)
  }
}
```

```r
subgroup_vars_for_grouping <- c(
  group_vars,
  subgroup_label_vars
)

output_data |>
  pivot_longer(
    cols = all_of(response_vars),
    names_to = "VARIABLE",
    values_to = "VALUE"
  ) |>
  mutate(
    VALUE_CODE = if_else(
      is.na(VALUE),
      "Missing",
      as.character(VALUE)
    ),

    VALUE_LABEL = purrr::map2_chr(
      VARIABLE,
      VALUE_CODE,
      lookup_label
    )
  ) |>
  group_by(
    across(
      all_of(
        c(
          subgroup_vars_for_grouping,
          "VARIABLE",
          "VALUE_CODE",
          "VALUE_LABEL"
        )
      )
    )
  ) |>
  summarise(
    N = n(),
    .groups = "drop"
  ) |>
  group_by(
    across(
```

```
        all_of(
          c(
            subgroup_vars_for_grouping,
            "VARIABLE"
          )
        )
      )
    ) |>
    mutate(
      PERCENT = round(100 * N / sum(N), 2)
    ) |>
    ungroup() |>
    arrange(
      across(all_of(subgroup_vars_for_grouping)),
      VARIABLE,
      VALUE_CODE
    )
}
```

show summaries for any variable

```
summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7), # filter out women 50 year
   ↪  and above to reduce missingness in pregnant and insurance response
  group_vars = c(
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "MARITAL", # change this variable to any you want
  variable_labels = variable_labels
)
```

```
# A tibble: 1,748 x 8
   STATE   REGION  YEAR VARIABLE VALUE_CODE VALUE_LABEL               N PERCENT
   <chr>   <chr>  <int> <chr>    <chr>      <chr>                 <int>   <dbl>
 1 Alabama South   2017 MARITAL  1          Married                 581   46
 2 Alabama South   2017 MARITAL  2          Divorced                159   12.6
 3 Alabama South   2017 MARITAL  3          Widowed                  15    1.19
 4 Alabama South   2017 MARITAL  4          Separated                67    5.3
```

```
 5 Alabama South   2017 MARITAL  5          Never married            394   31.2
 6 Alabama South   2017 MARITAL  6          Member of an unmarrie~    43    3.4
 7 Alabama South   2017 MARITAL  9          Refused                    4    0.32
 8 Alabama South   2019 MARITAL  1          Married                  571   44.1
 9 Alabama South   2019 MARITAL  2          Divorced                 169   13.1
10 Alabama South   2019 MARITAL  3          Widowed                   23    1.78
# i 1,738 more rows
```

Pregnancy-eligible range population

```
pregnancy_eligible_pop <- analysis_data_v2 |> filter(AGEG5YR < 7)
# categorical variables
vars_to_plot <- setdiff(
  names(pregnancy_eligible_pop),
  c("STATE", "YEAR", "post_ban", "DIVISION", "MENTHLTH", "REGION")
)

plot_list <- lapply(vars_to_plot, function(var) {

  # Convert the variable to a factor with numeric-looking levels
  pregnancy_eligible_pop[[paste0(var, "_factor")]] <-
    factor(pregnancy_eligible_pop[[var]])

  var_factor <- paste0(var, "_factor")
  label_map <- variable_labels[[var]]

  # Function for legend labels
  legend_labels <- function(x) {
    x <- as.character(x)

    if (is.null(label_map)) {
      return(x)
    }

    value_labels <- unname(label_map[x])

    # Show "code = label"; fall back to the code if no label exists
    value_labels[is.na(value_labels)] <- x[is.na(value_labels)]

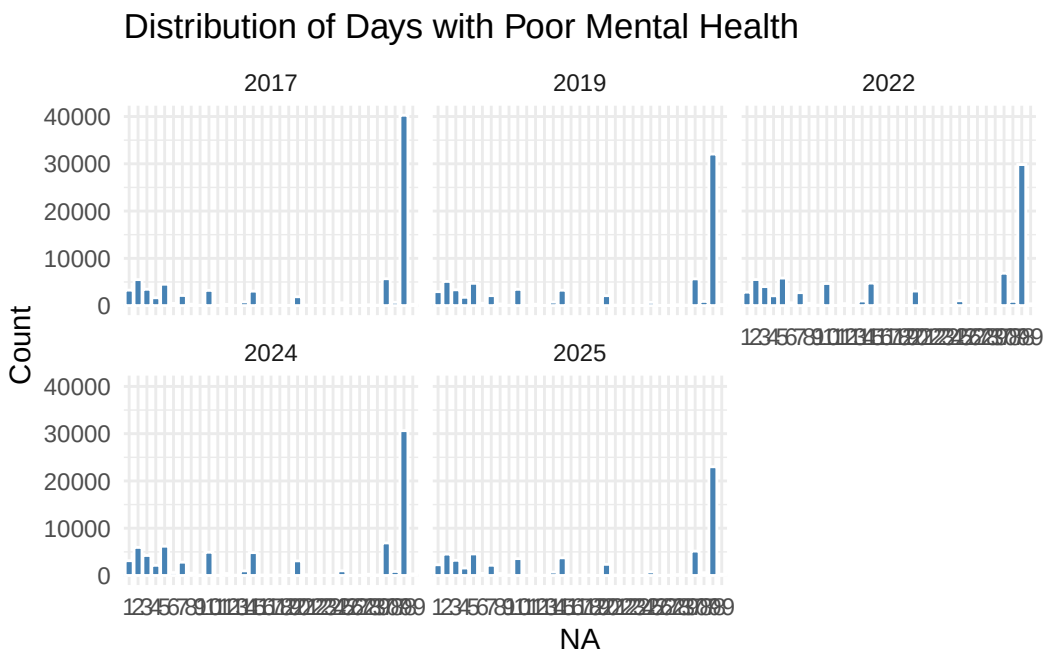
    paste0(x, " = ", value_labels)
  }
```

```
  ggplot(
    pregnancy_eligible_pop,
    aes(
      x = .data[[var_factor]],
      fill = .data[[var_factor]]
    )
  ) +
    geom_bar(color = "white") +
    facet_wrap(~YEAR) +
    scale_fill_viridis_d(
      name = var,
      labels = legend_labels,
  na.value = "grey70",
  na.translate = TRUE
    ) +
    theme_minimal() +
    theme(
      axis.text.x = element_text(
        angle = 45,
        hjust = 1
      ),
      legend.position = "right",
      legend.text = element_text(size = 9)
    ) +
    labs(
      x = var,
      y = "Count"
    )
})

# numerical variable
  ggplot(
  pregnancy_eligible_pop ,
  aes(x = factor(MENTHLTH))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
```

```
    title = "Distribution of Days with Poor Mental Health",
    x = var,
    y = "Count"
  )
```

```
Warning in label(labels[[nm]] %||% ""): NAs introduced by coercion
```




```
names(plot_list) <- vars_to_plot

# Choose the folder where plots should be saved
output_dir <- "plots"

# Create the folder if it does not already exist
dir.create(output_dir, showWarnings = FALSE, recursive = TRUE)

# Save every plot
for (plot_name in names(plot_list)) {

  ggsave(
    filename = file.path(output_dir, paste0(plot_name, ".png")),
    plot = plot_list[[plot_name]],
```

```
    width = 10,
    height = 7,
    units = "in",
    dpi = 300
  )
}
```

Trends in the U.S. Regions Over Time

```
plot_variable <- function(data, variable_name, plot_title, legend_title) {

  data |>
    dplyr::filter(VARIABLE == variable_name) |>
    ggplot(
      aes(
        x = factor(YEAR),
        y = PERCENT,
        fill = VALUE_LABEL
      )
    ) +
    geom_col(position = "dodge") +
    facet_wrap(~ REGION) +
    labs(
      title = plot_title,
      x = "Year",
      y = "Percent",
      fill = legend_title
    ) +
    theme_minimal() +
    theme(
      legend.position = "bottom",
      axis.text.x = element_text(angle = 45, hjust = 1)
    )
}
```

```
var_by_region_plots <- list(
  marital = plot_variable(
    summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7),
  group_vars = c(
```

```
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "MARITAL",
  variable_labels = variable_labels
),
    "MARITAL",
    "Marital Status by Region and Year",
    "Marital status"
  ),

  home = plot_variable(

    data = summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7),
  group_vars = c(
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "HOME",
  variable_labels = variable_labels
),
    "HOME",
    "Homeowner Status by Region and Year",
    "Homeowner status"
  ),

  pregnant = plot_variable(
    data = summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7),
  group_vars = c(
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "PREGNANT",
  variable_labels = variable_labels
),
    "PREGNANT",
    "Pregnancy Status by Region and Year",
    "Pregnancy status"
```

```
  ),

  employ = plot_variable(
    data = summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7),
  group_vars = c(
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "EMPLOY",
  variable_labels = variable_labels
),
    "EMPLOY",
    "Employment Status by Region and Year",
    "Employment status"
  ),

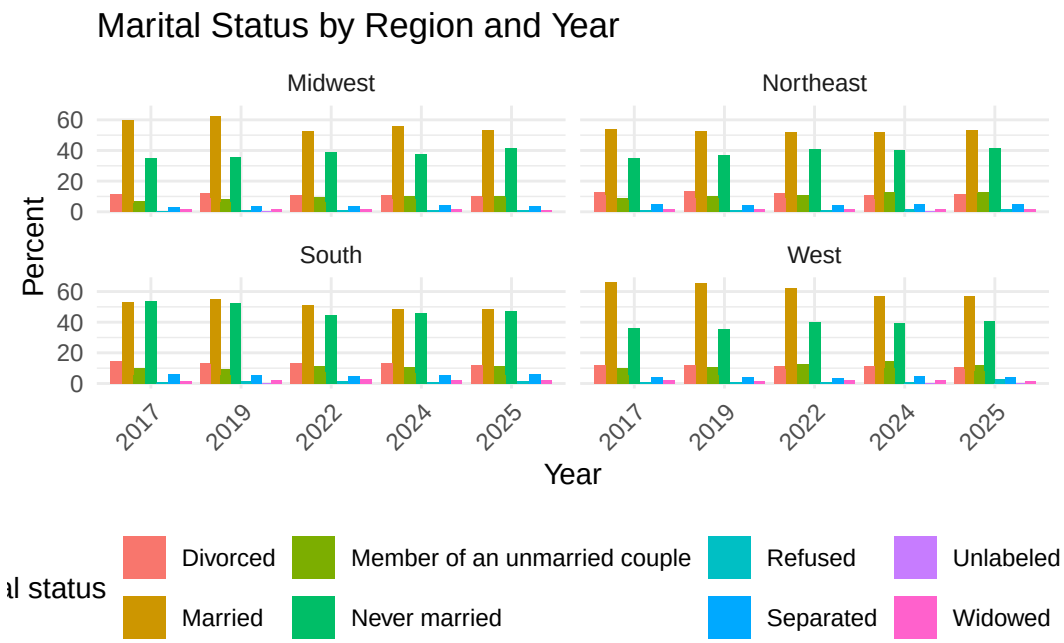
  income = plot_variable(
    data = summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7),
  group_vars = c(
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "INCOME",
  variable_labels = variable_labels
),
    "INCOME",
    "Income Categories by Region and Year",
    "Income category"
  ),

  insurance = plot_variable(
    data = summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7),
  group_vars = c(
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "INSURANCE",
```

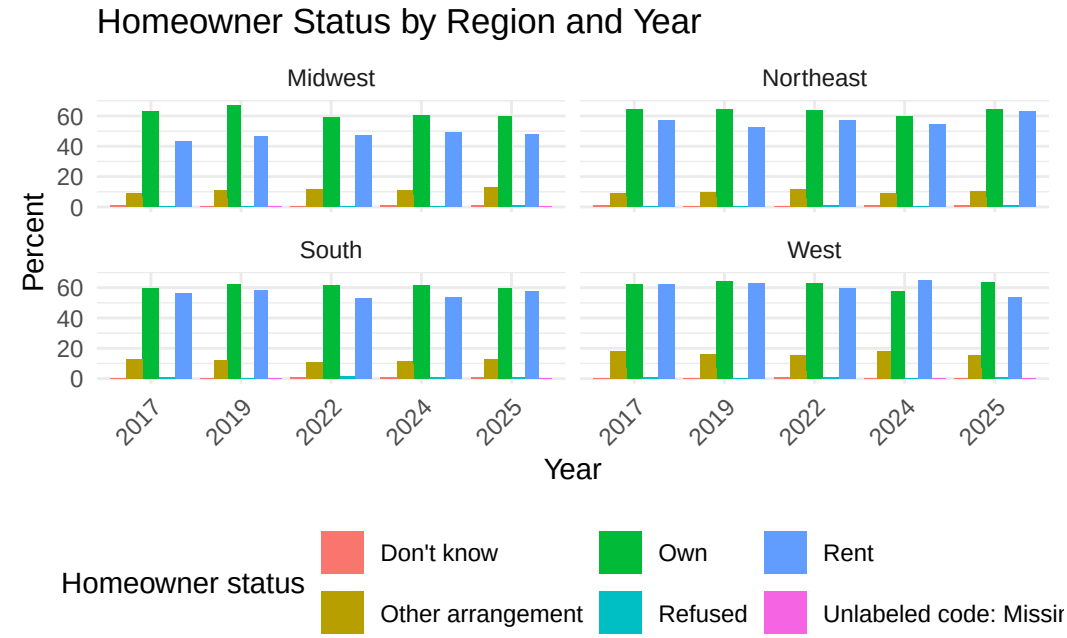
```
  variable_labels = variable_labels
),
    "INSURANCE",
    "Health Insurance Status by Region and Year",
    "Insurance status"
  ),

  race = plot_variable(
    data = summarize_categorical(
  data = analysis_data_v2 |> filter(AGEG5YR < 7),
  group_vars = c(
    "STATE",
    "REGION",
    "YEAR"
  ),
  response_vars = "RACE",
  variable_labels = variable_labels
),
    "RACE",
    "Race and Ethnicity by Region and Year",
    "Race and ethnicity"
  )
)
```

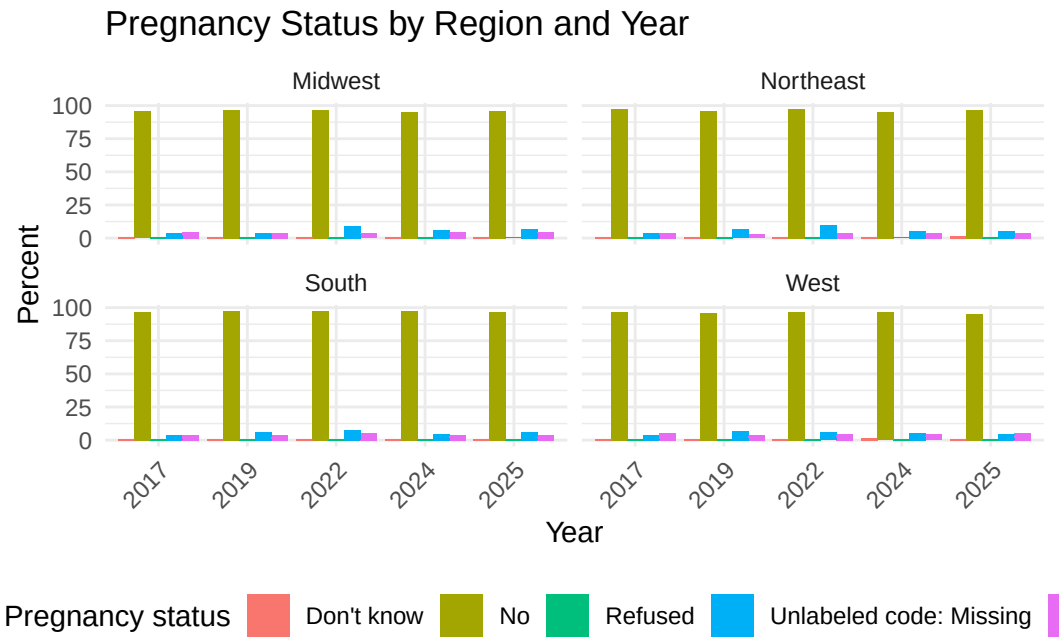
```
var_by_region_plots$marital
```



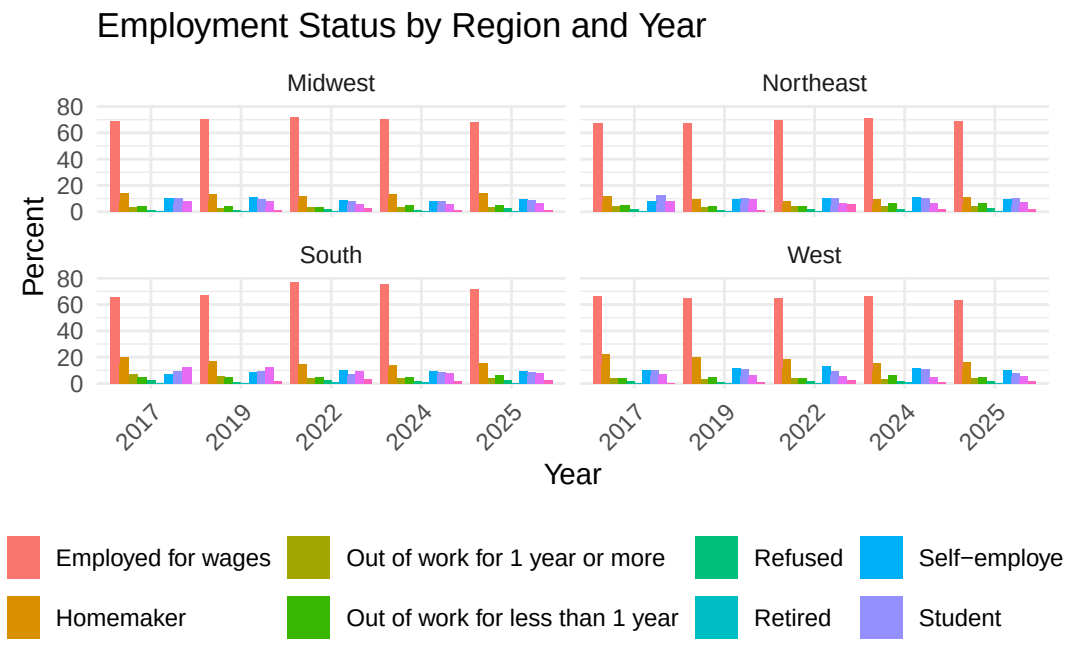

```
var_by_region_plots$home
```



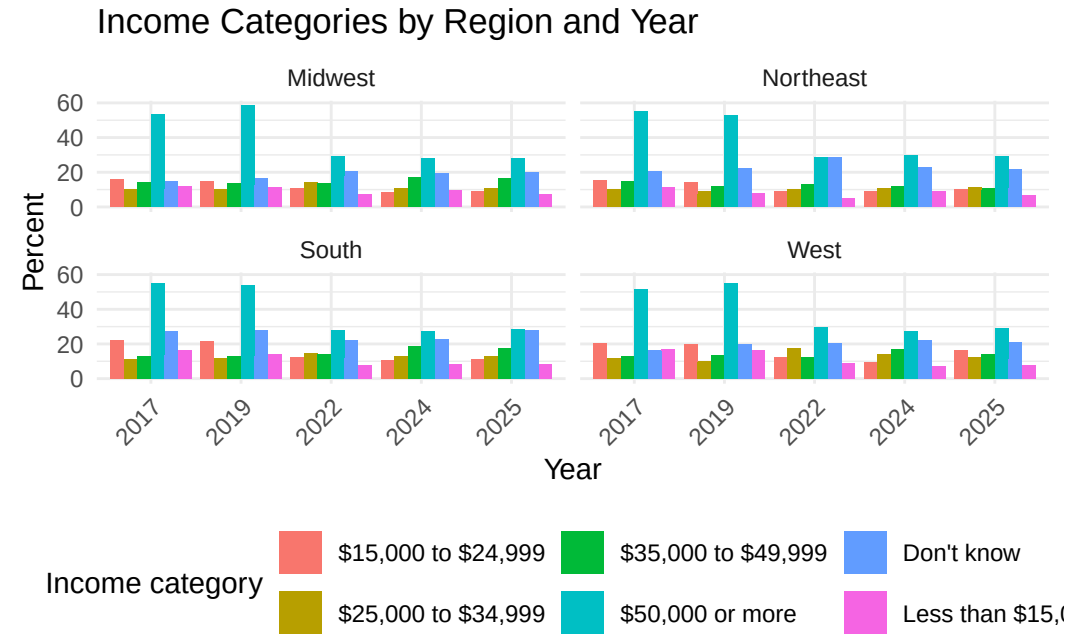
```
var_by_region_plots$pregnant
```



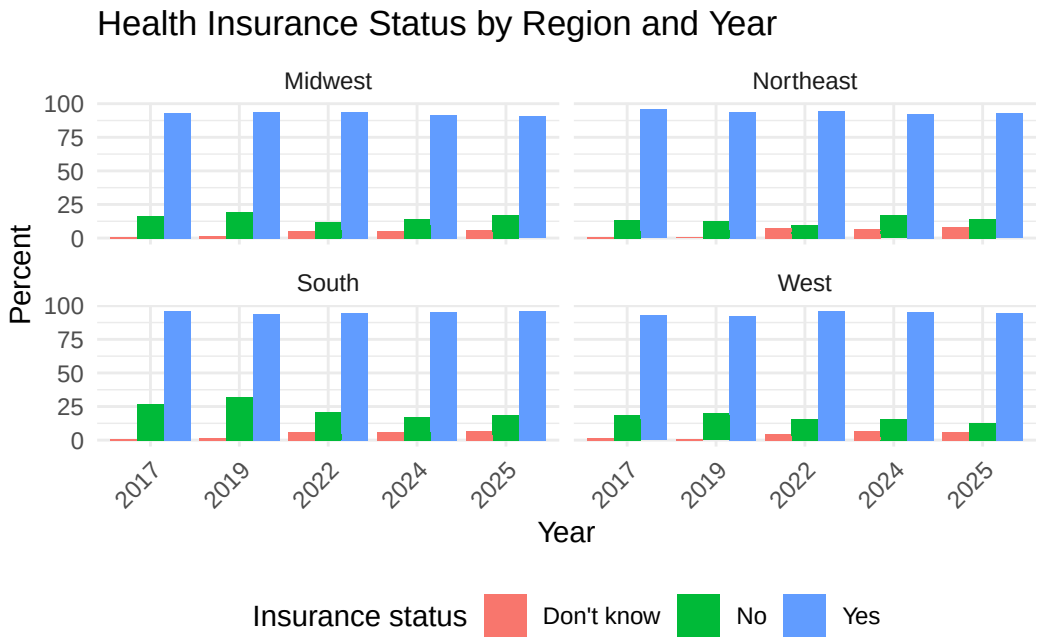

```
var_by_region_plots$employ
```



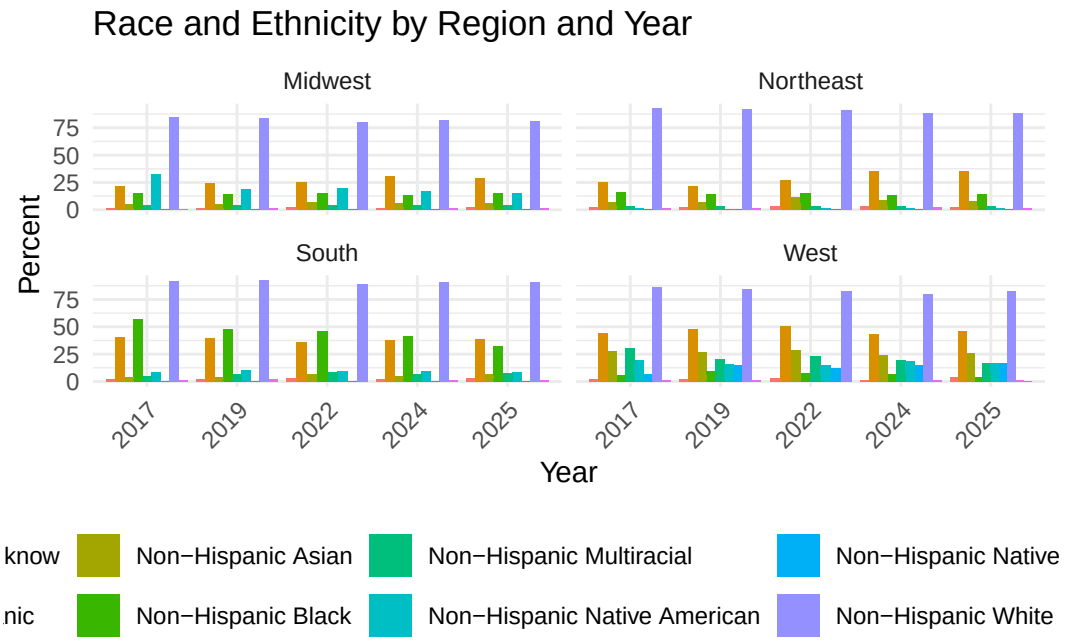

```
var_by_region_plots$income
```



```
var_by_region_plots$insurance
```




```
var_by_region_plots$race
```


Race and Ethnicity by Region and Year
Midwest
Northeast
South
West
Percent
75
50
25
0
2017
2019
2022
2024
2025
Year
know
nic
Non−Hispanic Asian
Non−Hispanic Black
Non−Hispanic Multiracial
Non−Hispanic Native American
Non−Hispanic Native
Non−Hispanic White